

(A) Local first-order mechanism

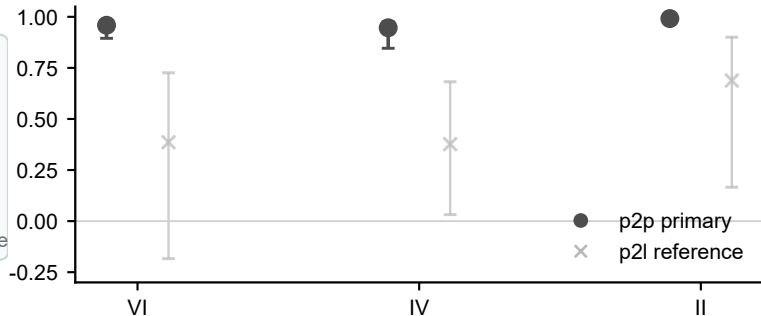
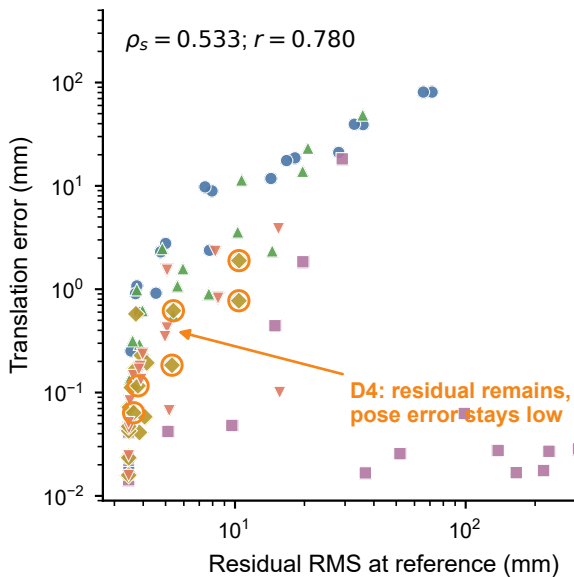
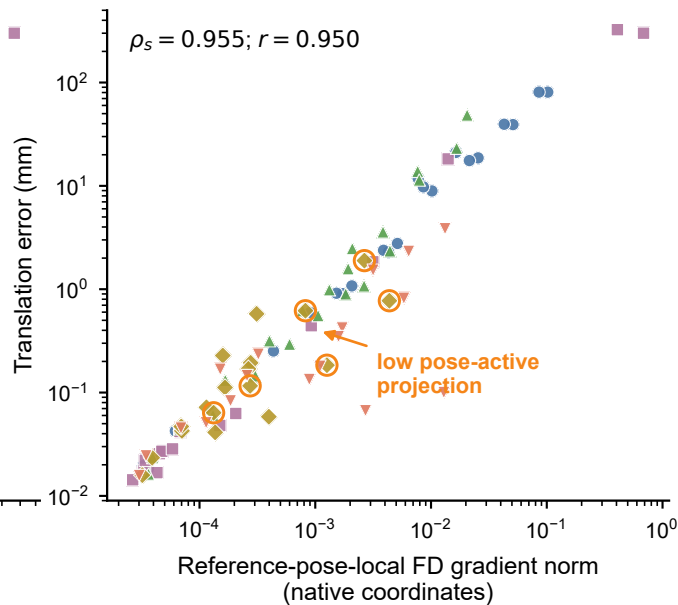
$$r(\Delta\xi) \approx \delta + J\Delta\xi$$

$$g_{\text{ref}} = J^T W \delta$$

$$\Delta\xi_{\text{FO}} = -H_{\text{GN}}^\dagger g_{\text{ref}}$$

fixed correspondences; observable subspace

Direction cosine

(B) Predicted versus realized translation**(C) Residual magnitude****(D) Pose-active gradient diagnostic**

● D1 ■ D2 ▲ D3 ◆ D4 ▼ D5